

PM 2.5	EST. AQI	WARMING	BY 2030	BY 2050
181.0	272	+0.51	25.7	25.9
ug/m3	index	deg C	deg C pred	deg C pred
CRITICAL	CRITICAL	SAFE	PROJECTION	PROJECTION

CLIMATE TREND ANALYSIS

Historical temperature records for **Pune** indicate a steady warming trajectory. Annual mean temperatures have increased by **+0.51 deg C** since the baseline period, at a rate of **0.0063 deg C per year**. Projections place the city at **25.7 deg C** by 2030 and **25.9 deg C** by 2050, underscoring the urgency of green infrastructure investment and urban heat mitigation.

Warming Rate	0.0063 deg C/yr
Total Warming	+0.51 deg C
2030 Projection	25.7 deg C
2050 Projection	25.9 deg C

AIR QUALITY & POLLUTION ASSESSMENT

Current PM2.5 for **Pune**: **181.0 ug/m3** — CRITICAL - Hazardous. WHO annual guideline: 15 ug/m3. Estimated AQI: **272**. Sustained exposure above 60 ug/m3 significantly elevates cardiovascular and respiratory disease risk across the urban population.

PRIORITY INTERVENTIONS & CIVIC ACTION PLAN

#	Initiative	Action Directive	Priority
P1	Air Quality Control	Implement alternate-day vehicle circulation, mandate industrial scrubbers, increase road-sweeping 3x.	CRITICAL
P2	Urban Cooling Initiative	Scale cool-roof painting across commercial zones. Target 40% reflective coverage by year-end.	HIGH
P3	Green Infrastructure	Plant 50,000 native trees along heat corridors. Mandate rooftop gardens for new builds > 500 m2.	HIGH

P4	Heatwave Protocol	Pre-position cooling centres with hydration stations. Issue alerts when LST > 44 C for two consecutive days.	MEDIUM
P5	EO Monitoring	Deploy automatic satellite anomaly alerts via MODIS/Sentinel. Weekly automated AQI briefings to council.	STANDARD